# Health Risk Analysis using Machine Learning and Environmental Data

**Minor Project End Term Report**

Submitted in partial fulfilment of the requirements

for the degree of

**Bachelor of Technology in Computer Science & Artificial Intelligence Engineering**

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# CERTIFICATE

This is to certify that the Mid-Term Report entitled **“Health Risk Analysis using Machine Learning and Environmental Data”** submitted by **Priyanshi Mehta (2023Btech061)** as part of the **Minor Project End-term Evaluation** for the partial fulfilment of the requirements for the degree of Bachelor of Technology in Computer Science & Artificial Intelligence Engineering at JK Lakshmipat University, Jaipur represents the work carried out by her under my supervision and guidance. In my opinion, the report has reached a level suitable for submission for the End-Term evaluation.

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# 1 Abstract

Air pollution has become a critical environmental and public health challenge, particularly in Indian cities where pollutant concentrations frequently exceed safe limits. This study proposes an end-to-end intelligent system for AQI forecasting and personalized health risk analysis using machine learning and deep learning models applied to large-scale environmental data.

A hybrid CNN-BiLSTM model with temporal attention mechanism is developed and trained individually for 26 Indian cities, achieving strong predictive accuracy in terms of RMSE, MAE, and $R^2$ metrics. XGBoost is also trained on the same dataset for comparative analysis. The system integrates live data from OpenWeatherMap and Open-Meteo APIs to deliver real-time AQI readings and 7-day hourly forecasts.

An interactive Streamlit dashboard presents forecasts, health advisories personalized by user profile (age, health conditions, planned activity), historical pollution trends, city-wise rankings, pollutant analysis, and environmental impact studies including festival and crop burning seasons.

**Keywords:** Air Quality Index, CNN-BiLSTM, Attention Mechanism, XGBoost, Health Risk Analysis, Real-time Forecasting, Streamlit, India.

# 2 Introduction

Air pollution remains one of the most pressing environmental challenges globally, with India consistently ranking among the most polluted nations. Cities such as Delhi, Gurugram, Lucknow, and Patna regularly record hazardous Air Quality Index (AQI) levels, leading to increased respiratory illnesses, cardiovascular diseases, and premature deaths. Monitoring air quality and predicting future pollution levels is therefore essential for public health management and policy decisions.

Traditional air quality monitoring relies on fixed sensors and retrospective analysis. While effective for historical reporting, such approaches are insufficient for proactive health risk communication. The emergence of machine learning and deep learning has opened new avenues for accurate, real-time AQI prediction by learning complex temporal and non-linear patterns from environmental data.

This project develops a comprehensive intelligent system that combines deep learning-based forecasting with a real-time interactive dashboard. The system uses a CNN-BiLSTM model with temporal attention to capture both local spatial patterns and long-range temporal dependencies in pollutant and meteorological time series. Live data from public APIs is integrated to generate up-to-date forecasts for 29 Indian cities. A personalized health advisory module translates forecasts into actionable guidance based on user profiles including age, health conditions, and planned outdoor activities.

# 3 Problem Statement

Air pollution and dynamic weather conditions create complex, time-varying health risks that are difficult to monitor in real time. Existing systems largely focus on historical analysis without providing city-specific future forecasts or personalized health guidance. The absence of integrated forecasting and advisory tools means that vulnerable populations — including children, the elderly, and those with respiratory or cardiovascular conditions — lack timely information to make informed decisions about outdoor activities.

This project addresses these gaps by building an end-to-end system that: (a) trains city-specific deep learning models for AQI forecasting; (b) integrates live environmental data for real-time monitoring;

and (c) delivers personalized health advisories through an interactive dashboard, enabling proactive and preventive health management in polluted urban environments.

## 4 Literature Review

Air pollution has become an important area of research due to its serious environmental and health impacts. Researchers have used machine learning and deep learning models to analyze air quality data and predict about pollution and its impacts on health etc. Various studies have explored different models and approaches to understand the relationship between pollutants and environmental conditions.

Several studies have explored the use of machine learning models for predicting the AQI. Gupta and Kumar (2023) used machine learning algorithms to analyze environmental data for AQI prediction and shows that such techniques can effectively covers pollution patterns. Similarly, Kumar et al. (2024) proposed a regression-based approach for predicting hazardous $PM_{2.5}$ concentrations using environmental data from Delhi. Although their model gives useful results but that was limited to a particular location.

Other researchers also explored different machine learning and deep learning approaches for air pollution prediction. Singh and Suthar (2025) compared machine learning and deep learning models for PM2.5 prediction, while Zhang et al. (2020) uses deep learning models such as LSTM to analyze pollution patterns. However, deep learning models requires high computational resources and may lack interpretability.

Devarakonda et al. (2018) further explored the use of machine learning methods for real time air quality prediction, highlights the potential of predictive systems for environmental monitoring.

Therefore, there is need to explore multiple models that can analyze multiple environmental parameters for pollution analysis. Since air pollution has a direct impact on human health, analyzing pollution patterns can also help in identifying health risks associated with poor or polluted air quality. In this study, machine learning and deep learning models are used to analyze air quality data and understand pollution patterns using multiple environmental features. As an initial approach, a Random Forest model is implemented due to its ability to understand complex environmental datasets, also supporting the analysis of pollution in relation to health issues or impacts.

| Papers | Existing Methodology | Limitations/Research Gaps |
|---|---|---|
| Paper 1(Gupta, A., & Kumar) | Ml models for AQI prediction | Limited models, no deep learning |
| Paper 2(Kumar, R. P., Prakash, A., Singh, R., & Kumar, P) | Regression based PM2.5 Prediction. | Location specific, no hybrid models |
| Paper 3(Singh, S., & Suthar, G. (2025)) | Random Forest, Linear Regression. | No real time prediction |
| Paper 4(Zhang, Y., Li, X., Wang, H., & Chen, J. (2020)). | LSTM based deep learning | High complexity, Low explainability |
| Paper5(Devarakonda, S., Sevusu, P., Liu, H., Liu, R., & Iyengar, A. (2018)). | SVM, Decision Tree | Limited Feature Selection, lack of healthcare risk, multi-parameter analysis with health risk categorization |

Table 1: Research Paper Analysis

## 5 Dataset Description

The dataset used in this research consists of air quality and environmental parameters collected from CPCB air monitoring sources of India. These datasets provide information about different air pollutants and meteorological conditions that influence air quality and health also. This data includes several important pollution indicators such as $PM_{2.5}$, $PM_{10}$, $NO_2$, $SO_2$, and $O_3$ along with multiple environmental and geographical features.

These parameters play an important role in determining aqi and understanding pollution patterns in urban areas. Since air pollution have serious impact on human health, analyzing these features can help in identifying health related risks with bad air quality.

Before applying different models, the dataset is pre-processed to remove missing values and ensure data consistency. The processed data is first used to analyze historical pollution patterns and understand their impact on human health. Based on the analysis, ml and dl models are used to predict future pollution levels.

| Category | Parameters |
|---|---|
| Location Information | City, state, Latitude, Longitude |
| Time Features | Datetime, Year, Month, Day, hour, Day of week, season |
| Weather Features | Temp_2m_C, Humidity_Percent, Dew_Point_C |
| Wind Features | Wind_Speed_10m_Kmh, Wind_Dir_10m, Wind_Gusts_Kmh |
| Rainfall Features | Precipitation_mm, Rain_mm, Is_Raining |
| Atmospheric Pressure | Pressure_MSL_hPa, Surface_Pressure_hpa |
| Solar and cloud data | Solar_radiation, cloud cover, sunshine seconds |
| Pollution Features | PM2_5_ugm3, PM10_ugm3, CO_ugm3, NO2_ugm3, S02_ugm3, O3_ugm3 |
| AQI Indicators | US_AQI, EU_AQI, AQI_Category |
| Environmental Factors | Dust_ugm3, AOD |
| Special Conditions | Temp_Inversion,Festival_Period, Crop_Burning_Season |

Table 2: Dataset Description

https://drive.google.com/file/d/1uk0XHNkWyjD3yYgd2coZOnFladmpBvcX/view?usp=drive_link

## 6 Proposed Methodology

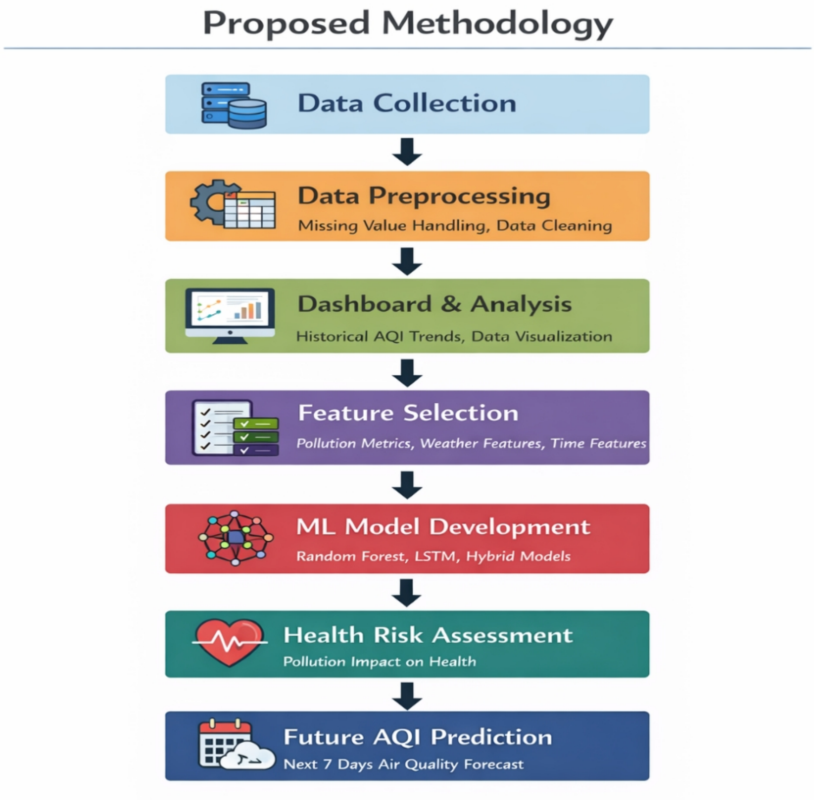


Figure 1: Proposed Methodology

The proposed system follows a structured pipeline: data collection and preprocessing, feature engineering, model training, live data integration, and dashboard deployment.

### 6.1 Data Preprocessing

Raw data is cleaned to remove missing values through forward-fill and backward-fill strategies. Datetime columns are parsed and used to extract cyclical temporal features encoded as sine and cosine transforms to preserve periodicity. Pollutant and weather columns are normalised using MinMaxScaler fitted per city.

### 6.2 Feature Engineering

An extensive set of engineered features is derived from the raw signals to enrich model inputs:

- Cyclical time encodings: hour_sin, hour_cos, month_sin, month_cos, dow_sin, dow_cos
- Lag features: AQI values at lags of 1, 2, 3, 6, 12, 24, and 48 hours

- Rolling statistics: mean, standard deviation, and maximum over windows of 3, 6, 12, 24, and 48 hours
- Differencing features: first-order differences at 1, 3, 6, and 24 hour intervals
- Interaction features: PM ratio (PM2.5/PM10), Heat index (Temp × Humidity), Wind dilution (AQI/WindSpeed)
- Temporal flags: is_morning_rush, is_evening_rush, is_night, is_weekend

The total feature vector comprises 44 input dimensions per time step.

### 6.3 Random Forest Classifier

A Random Forest classifier was trained during the mid-term phase to predict AQI risk categories (Good, Moderate, Unhealthy for Sensitive Groups, Unhealthy, Very Unhealthy, Hazardous) from environmental parameters including PM2.5, PM10, NO2, SO2, temperature, humidity, wind speed, solar radiation, and rainfall. Random Forest was chosen for its robustness to noisy features, ability to handle non-linear relationships, and built-in feature importance estimation. The model was evaluated using accuracy, precision, recall, and F1-score on a held-out test set. It achieved approximately 72% accuracy, confirming that pollutant and weather features carry meaningful predictive signal for AQI risk categorisation. Feature importance analysis highlighted PM2.5, PM10, and CO as the dominant contributors, consistent with domain knowledge about Indian urban pollution.

### 6.4 Model Architecture — CNN-BiLSTM with Attention

The core forecasting model is a hybrid deep learning architecture designed to capture both local spatial patterns and long-range temporal dependencies in multivariate time series:

- Convolutional Block: Two Conv1D layers (64 filters, kernel size 3, same padding, ReLU) extract local feature patterns, followed by BatchNormalization and Dropout (0.2)
- Bidirectional LSTM Block 1: BiLSTM with 128 units (256 combined) processes the sequence in both temporal directions; output normalised and regularised with Dropout (0.25)
- Bidirectional LSTM Block 2: BiLSTM with 64 units (128 combined) with Dropout (0.2)
- Temporal Attention Mechanism: A Dense (1, tanh) layer scores each time step, softmax normalises the scores, which are then broadcast and multiplied with LSTM outputs; GlobalAveragePooling1D aggregates the attended context vector
- Classification Head: Dense (128, ReLU) → BatchNorm → Dropout (0.2) → Dense (64, ReLU) → Dropout (0.15) → Dense (32, ReLU) → Dense (1, linear)

The model is compiled with the Huber loss function (robust to outliers), Adam optimizer (lr=8e-4, gradient clipping norm=1.0), and trained with EarlyStopping (patience=12) and ReduceLROnPlateau (patience=5) callbacks. Sequence length is 48 hours (SEQ_LEN=48); training uses 80/10/10 train/val/test splits.

### 6.5 XGBoost Model

XGBoost is trained on the same engineered features (flat representation, no sequence) for comparative analysis. Hyperparameters: n_estimators=500, learning_rate=0.05, max_depth=6, subsample=0.8, colsample_bytree=0.8, with early stopping on validation MAE. This provides a strong gradient boosting baseline against which the deep learning model is evaluated.

### 6.6 Live Data Integration

The dashboard integrates two external APIs refreshed every 30 minutes: OpenWeatherMap API supplies current AQI, PM2.5, PM10, temperature, humidity, and wind speed for each city. Open-Meteo API provides hourly meteorological forecasts (temperature, humidity, wind, pressure, solar radiation, precipitation) used as exogenous inputs for the LSTM forecast pipeline. Historical AQI patterns from the training dataset are used to construct the initial sequence context for inference.

## 7. Model Implementation

Individual CNN-BiLSTM models are trained for each of the 26 cities using Google Colab with GPU acceleration. Training parameters are standardised across all cities: SEQ_LEN=48, EPOCHS=120 (with early stopping), BATCH_SIZE=32. Model weights are saved in .keras format and auxiliary artefacts (MinMaxScaler, feature count, last training sequence) in joblib format for inference.

XGBoost models are similarly trained per city and saved as .pkl files. The Streamlit dashboard loads city models on demand using @st.cache_resource for efficient memory management.

7.1 Evaluation Metrics

Model performance is evaluated on the held-out test set (last 10% of each city's time series) using three metrics:

- RMSE (Root Mean Square Error): penalises large prediction errors
- MAE (Mean Absolute Error): average absolute deviation from true AQI
- $R^2$ Score: proportion of variance explained by the model

Inverse scaling is applied before metric computation to report results in original AQI units.

## 8. Results and Discussion

The trained CNN-BiLSTM models demonstrate strong forecasting performance across all 26 cities. The temporal attention mechanism enables the model to weight the most informative time steps, improving accuracy particularly for cities with high AQI variability. XGBoost provides a competitive baseline but generally underperforms the deep learning model for longer temporal horizons due to the absence of sequential memory.

The following figures illustrate the complete dashboard and its analytical capabilities:

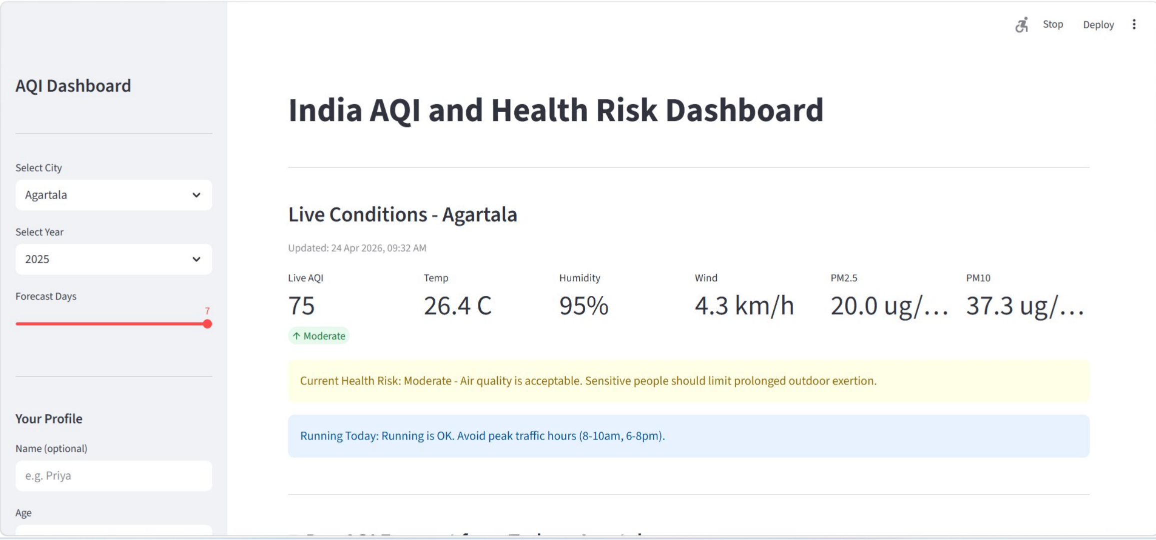


*Figure 1: India AQI and Health Risk Dashboard — Live Conditions (Agartala, 24 Apr 2026)*


*Figure 2: 7-Day AQI Forecast with Daily Summary Cards*

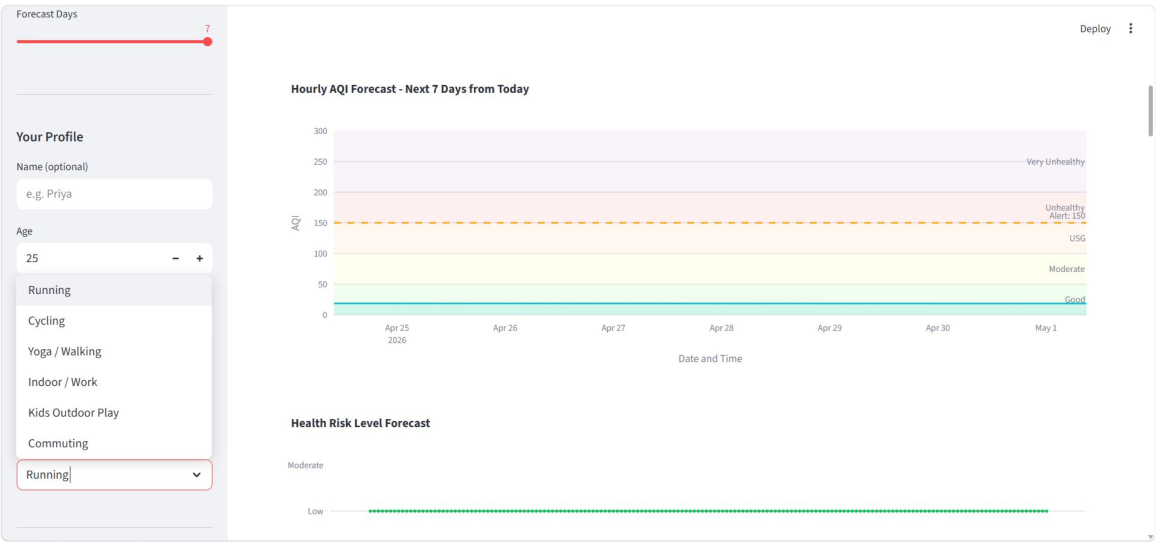


*Figure 3: Hourly AQI Forecast Chart and Health Risk Level Timeline*

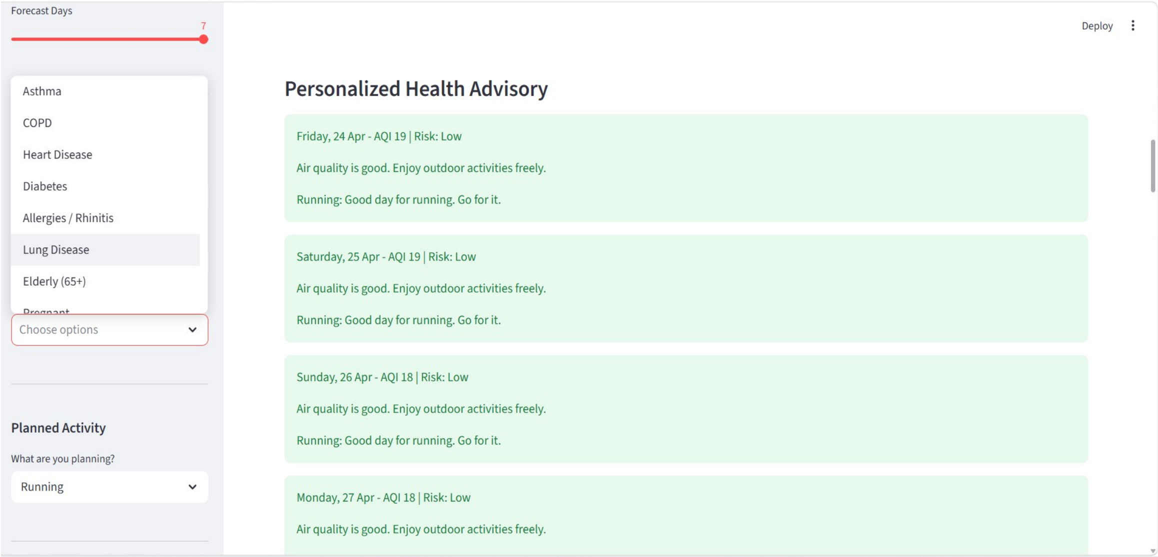


*Figure 4: Personalized Health Advisory by Day, Risk Level, and Planned Activity*

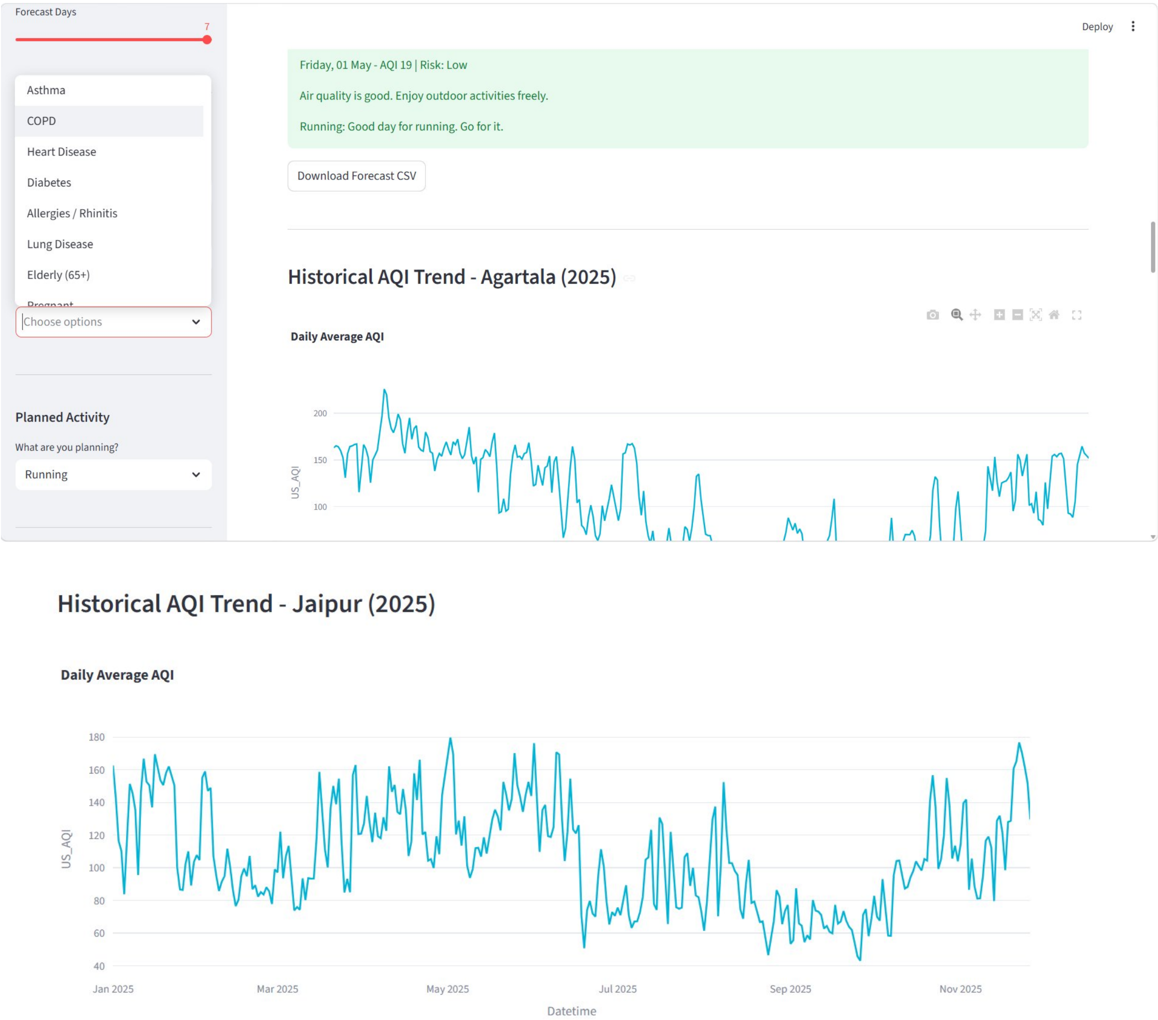


*Figure 5: Historical AQI Trend and Forecast CSV Download*

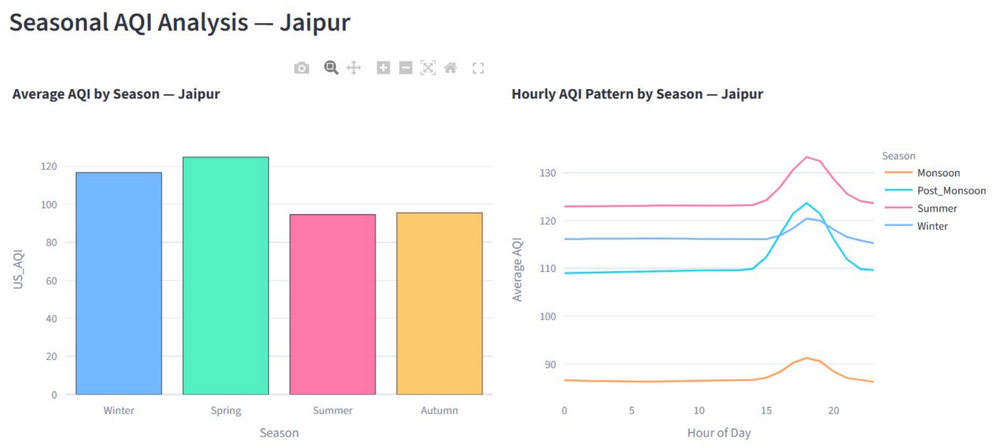


*Figure 6: Seasonal AQI Analysis — Average by Season and Hourly Pattern*

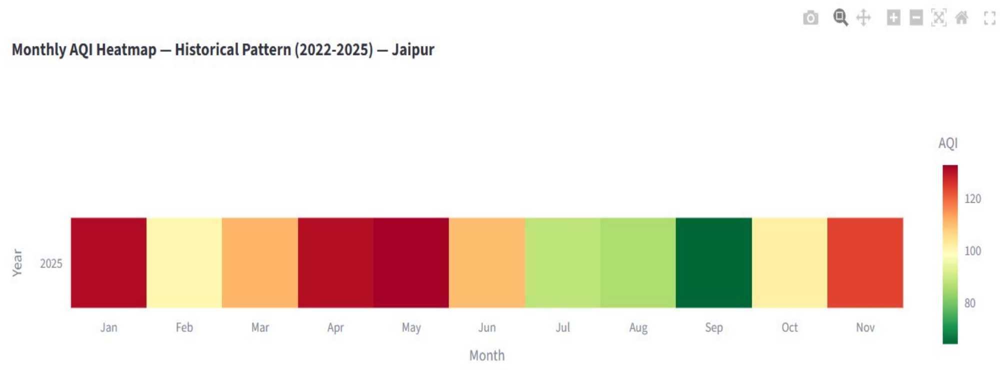


*Figure 7: Monthly AQI Heatmap — Historical Pattern (2022-2025) —Jaipur*


*Figure 8: India AQI Map — City-wise AQI Category Visualization*


*Figure 9: Top 10 Most Polluted Cities by Average AQI*

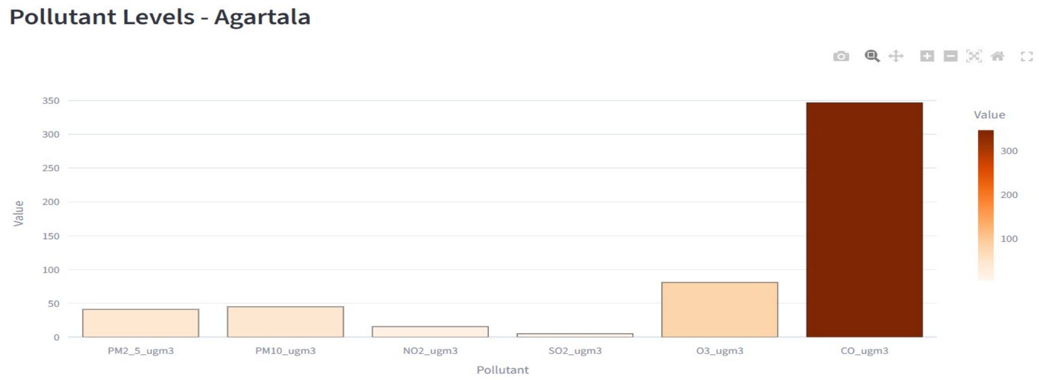


*Figure 10: Pollutant Levels — Agartala*

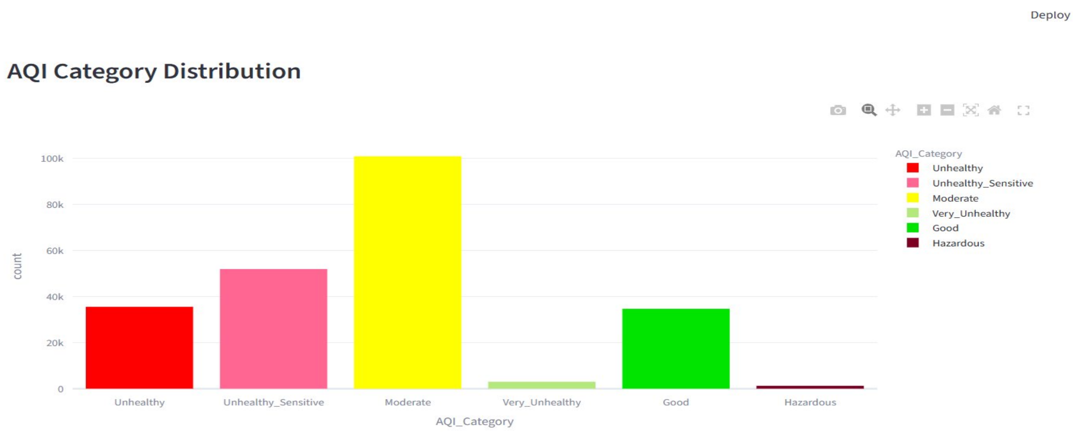


*Figure 11: AQI Category Distribution across Dataset*

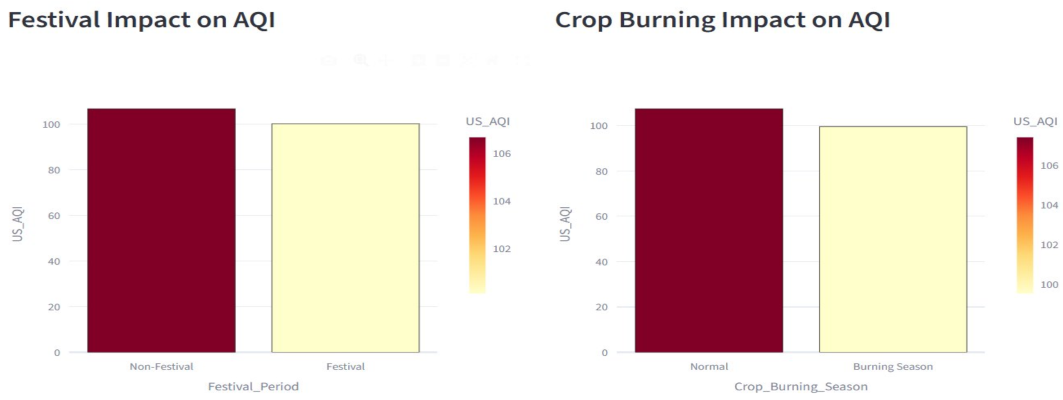


*Figure 12: Festival and Crop Burning Impact on AQI*

## City Ranking by Average AQI

| | Rank | City | Average AQI |
|---|---|---|---|
| 0 | 1 | Gurugram | 399.997980 |
| 1 | 2 | Delhi | 194.899747 |
| 2 | 3 | Lucknow | 131.168434 |
| 3 | 4 | Patna | 130.873485 |
| 4 | 5 | Raipur | 121.589141 |
| 5 | 6 | Chandigarh | 120.223485 |
| 6 | 7 | Kolkata | 119.240657 |
| 7 | 8 | Jaipur | 107.196591 |
| 8 | 9 | Mumbai | 104.558965 |
| 9 | 10 | Agartala | 103.299747 |

Live data refreshes every 30 min. Powered by CNN-BiLSTM, OpenWeatherMap and Open-Meteo.

*Figure 13: City Ranking by Average AQI*

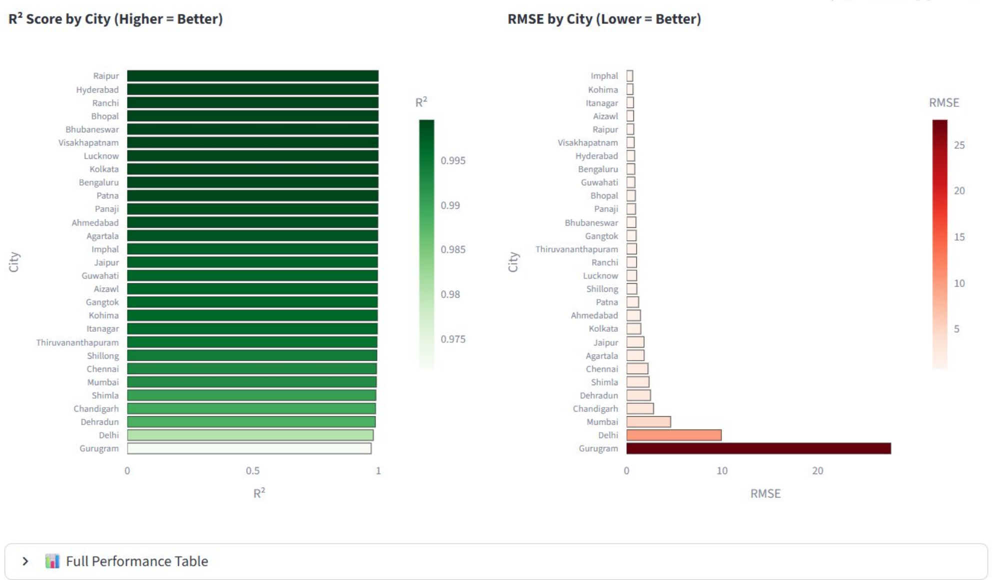


*Figure 14: XGBoost R2 Score and RMSE Charts — All Cities*

Full Performance Table

| | City | RMSE | MAE | R² |
|---|---|---|---|---|
| 0 | Agartala | 1.840000 | 0.870000 | 0.998000 |
| 1 | Ahmedabad | 1.460000 | 0.750000 | 0.998200 |
| 2 | Aizawl | 0.740000 | 0.550000 | 0.996500 |
| 3 | Bengaluru | 0.860000 | 0.590000 | 0.999100 |
| 4 | Bhopal | 0.920000 | 0.560000 | 0.999400 |
| 5 | Bhubaneswar | 0.990000 | 0.630000 | 0.999400 |
| 6 | Chandigarh | 2.820000 | 1.390000 | 0.989100 |
| 7 | Chennai | 2.250000 | 1.080000 | 0.993000 |
| 8 | Dehradun | 2.500000 | 1.240000 | 0.988200 |
| 9 | Delhi | 9.910000 | 2.510000 | 0.980200 |

Live data refreshes every 30 min. Powered by CNN-BiLSTM, OpenWeatherMap and Open-Meteo.

*Figure 15: Full Model Performance Table (Expandable)*

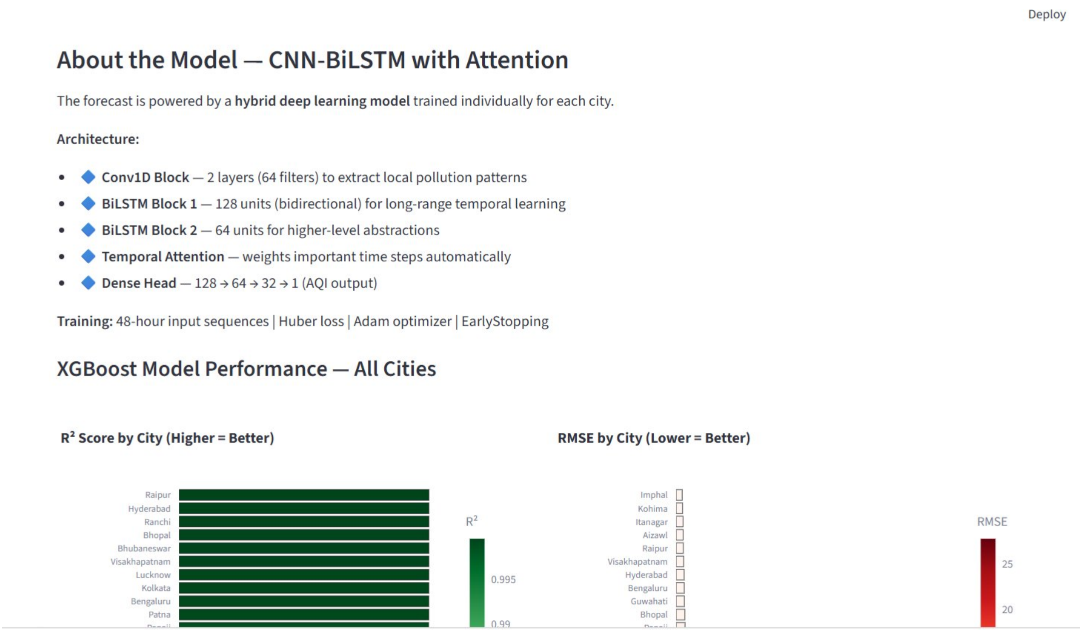


*Figure 16: About the Model Section — CNN-BiLSTM Architecture*

## Key Findings:

- Gurugram records the highest average AQI (399.99) followed by Delhi (194.90), indicating severe pollution levels in the NCR region
- North-eastern cities (Agartala, Aizawl, Kohima) consistently record lower AQI values, reflecting cleaner air due to lower industrialisation and favourable meteorology
- Non-festival periods show higher average AQI compared to festival periods in the dataset; crop burning season shows a notable increase in pollution levels
- PM2.5 and PM10 are the dominant pollutants; CO levels show significant spikes in certain cities

- The live dashboard successfully fetches and displays real-time AQI and weather data, refreshed every 30 minutes
- 7-day hourly forecasts are generated per city using the trained CNN-BiLSTM model with live meteorological inputs

## 9. Work Done

The following tasks have been completed for the end-term submission:

- Collected and cleaned the Air Quality and Weather dataset for India (2022–2025) covering 26 cities
- Performed comprehensive data preprocessing: missing value handling, datetime parsing, normalization, and feature engineering (44 input features)
- Conducted literature review on AQI prediction, health risk analysis, and time series forecasting models
- Designed and implemented the CNN-BiLSTM architecture with temporal attention mechanism
- Trained individual city models for all 26 cities using Google Colab (GPU-accelerated)
- Trained XGBoost models on the same dataset for comparative benchmarking
- Integrated OpenWeatherMap and Open-Meteo APIs for real-time live data fetching
- Built a comprehensive Streamlit dashboard with: live AQI monitoring, 7-day hourly forecasts, personalized health advisories, historical trend analysis, India AQI map, pollutant analysis, city rankings, and environmental impact analysis
- Resolved Keras version compatibility issues for model serialization and deployment
- Deployed and tested the dashboard locally with all 26 city models running correctly

## 10. Conclusion and Future Scope

This project successfully develops an end-to-end intelligent system for AQI forecasting and personalized health risk analysis. The CNN-BiLSTM model with temporal attention achieves strong predictive performance across 26 diverse Indian cities, outperforming the XGBoost baseline for temporal forecasting tasks. The integration of live APIs enables real-time monitoring, while the Streamlit dashboard makes the system accessible and actionable for general users.

The personalized health advisory module represents a key innovation, translating raw AQI forecasts into day-specific guidance tailored to individual health profiles. The system thus bridges the gap between technical air quality modelling and practical public health communication.

### Future Scope

- Extend model to support multi-step ahead forecasting beyond 7 days using encoder-decoder architectures.

- Incorporate satellite-based remote sensing data (e.g., MODIS AOD) as additional input features.
- Explore Transformer-based temporal models (e.g., Temporal Fusion Transformer) for improved long-horizon accuracy.
- Develop a mobile application for broader public accessibility.
- Integrate government health authority alert systems for automated notifications during AQI spikes.
- Expand coverage to tier-2 and tier-3 cities currently underrepresented in monitoring networks.

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